



AIxIA 2026

24th Int Italian Conference on AI



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The Impact of Domain-Specific Knowledge in Legal AI: A Case Study on Italian Criminal Law

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Context: Legal Qualification

Marco attacks a stranger outside a bar and kicks him repeatedly in the knee. The knee is fractured and, once healed, the victim is left with a permanent limp.

Facts of the case



"lesioni personali gravi per indebolimento permanente di un organo"

Rubrica Imputativa as NL



Art. 582(1) c.p. → *offense*: personal injury,

Art. 583(1)(2) c.p. → *circumstance*: the injury becomes "serious" when it permanently weakens a sense or an organ

Rubrica Imputativa as a set of violated provisions



- 3–7 years' imprisonment
- Pre-trial custody admissible
-

Legal consequences



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Context: The Winfried Platform

The screenshot displays the Winfried platform interface. On the left, a sidebar contains the 'winfried' logo and a navigation menu with sections: 'Rubrica imputativa', 'NORME INCRIMINATORIE' (listing art. 56 comma primo c.p. and art. 582 comma primo c.p.), and 'CIRCOSTANZE AGGRAVANTI' (listing art. 585 comma primo (1° periodo) c.p. and art. 576 comma primo num. 5) c.p.). The main content area is titled 'RITO ORDINARIO' and shows a list of legal provisions under the heading 'MISURE PRECAUTELARI'. The first section is 'Arresto', stating 'L'arresto non è consentito.' and providing 'Riferimenti normativi' (art. 56 comma secondo c.p.; art. 278 comma 1 c.p.p.; art. 380 c.p.p.; art. 381 c.p.p.) and 'Precedenti in senso conforme' (Corte di cassazione, sez. V, 26 novembre 2021 (dep. 29 marzo 2022), n. 11569, Feraru; Corte di cassazione, sez. II, 16 novembre 2010 (dep. 2 marzo 2011), n. 7995, Cassia). The second section is 'Fermo di indiziato di delitto', stating 'Il fermo di indiziato di delitto non è consentito.' and providing similar 'Riferimenti normativi' and 'Precedenti in senso conforme'.

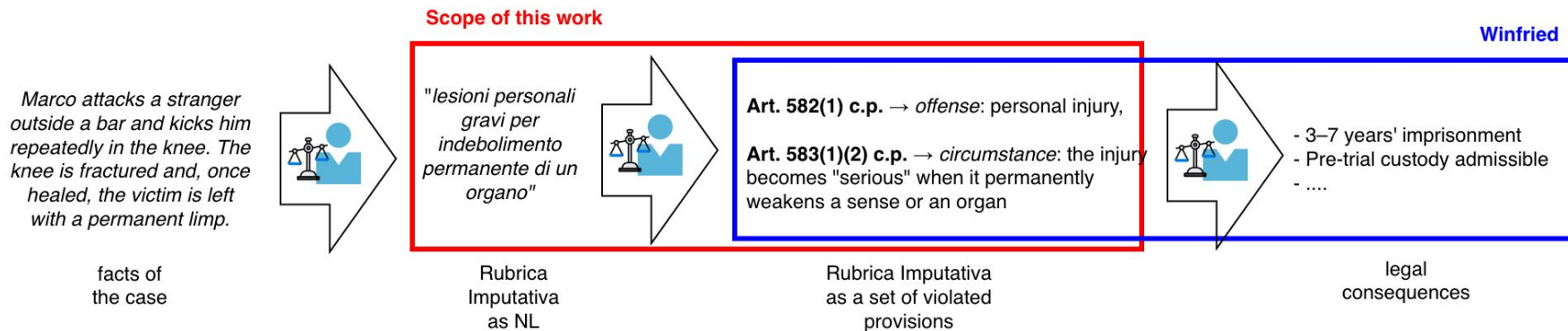
In production for defence counsel, judges and prosecutors

Curated KB: every provision typed as **offense** or **circumstance**; valid combinations mapped

Input: the exact articles, i.e. what the user is looking for



Problem Formulation: Scope





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Problem Formulation: Statutory Charge Retrieval

Given a natural-language query q , retrieve the complete list $R_q = (p_1, \dots, p_k)$ of provisions that constitute its rubrica imputativa (i.e. the charge)



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Flat queries: maps to a independent list of provisions

“omicidio” → intentional (Art. 575)
→ consensual (Art. 579)
→ preterintencional (Art. 584)
→ negligent omodice (Art. 589)



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Block-structured queries: specify a charge that exists only as a joint configuration of provisions

“truffa aggravate dalla minorata difesa”
→ fraud (Art. 640(1))
→ aggravating circumstances (Art. 640(2)(2-ter))



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Why is this task hard?

2,040

offense/circumstance
provisions



≈ 2.1 M

possible charges with 2
provisions



≈ 1.4 B

possible charges with 3
provisions

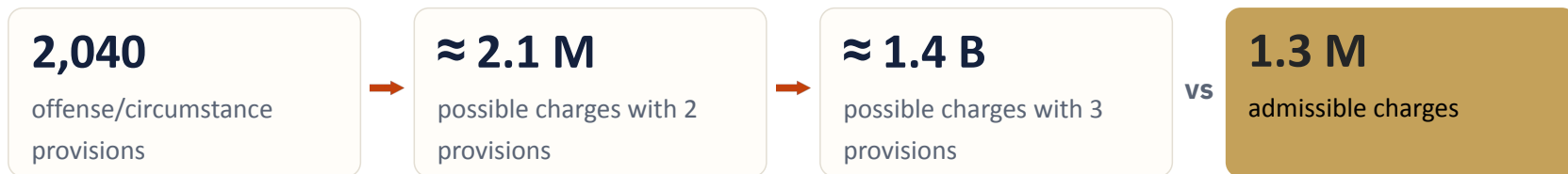
vs

1.3 M

admissible charges



Why is this task hard?



Query: *“lesioni personali gravi per indebolimento permanente di un organo”*

Art. 590-bis c.p.	<i>“lesioni personali stradali gravi...”</i>	negligent road-traffic injury
Art. 583-quater c.p.	<i>“lesioni gravi o gravissime”</i>	injury to police or health staff
Art. 572(3) c.p.	<i>“lesione personale grave”</i>	maltreatment of family members
Art. 583-quinquies c.p.	<i>“lesioni permanenti al viso”</i>	permanent facial disfigurement



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Intuition: Injecting Domain Knowledge in Retrieval

We argue that well-structured domain knowledge can guide retrieval in ways that semantic similarity alone cannot.



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Compatibility mapping:

for each offense, the circumstances the code admits.

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→ *recidivism*
→ *victim's vulnerability*
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combinations of provisions instead of a plain list of provisions with billions of possible combinations



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Provided by
Winfried



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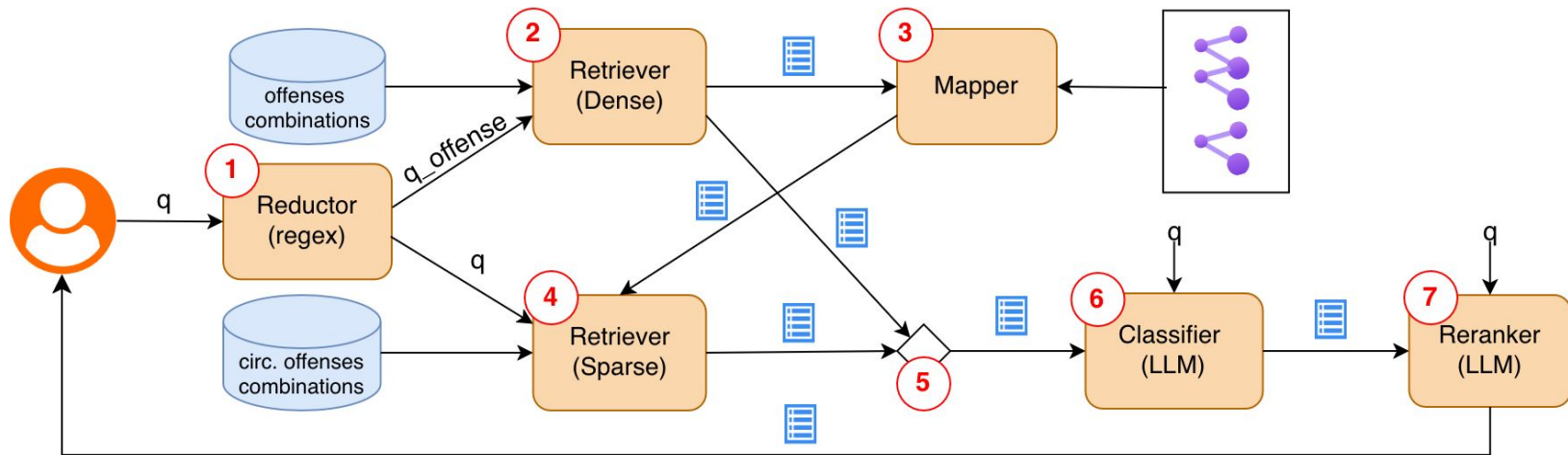
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Research Question: Alongside LLMs, does injecting domain knowledge still add value?

Knowledge Guided Multi-Stage Retrieval (KMSR)

KMSR is a pipeline built around the law's compositional structure, made explicit as knowledge





Experimental Setup

Dataset

- 350 real queries from paying users
- Annotated by legal experts
- 120 flat, 227 block-structured

Models

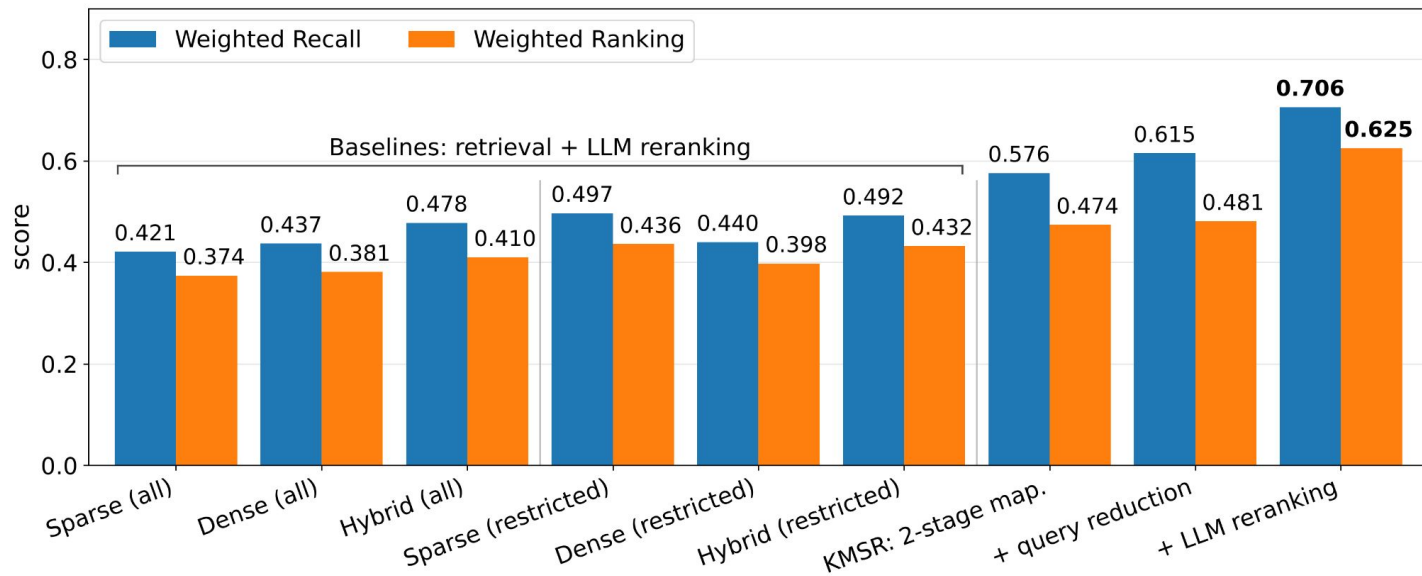
- Baselines = retriever + LLM reranker
 - Retrievers: sparse, dense, hybrid
 - Target corpus: all passages (10.784) or restricted (2.040)
- KMSR: 2-stages mapping, + q. reduct. + LLM reranker

Metrics

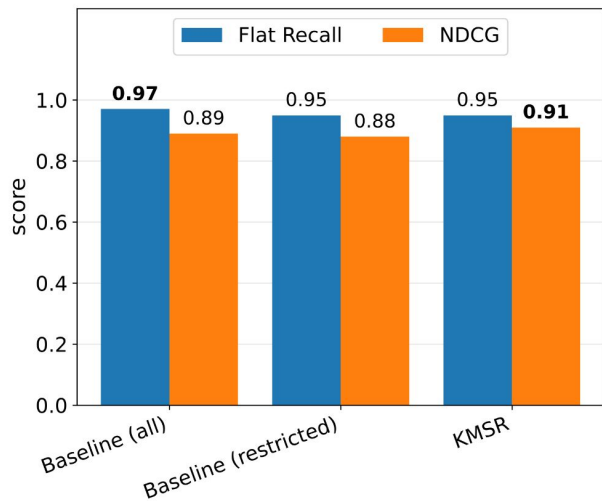
- Flat queries → Recall, NDCG
- Block queries → Block Recall, Block AP
- Mixed metrics → Weighted Recall, Weighted Ranking

GPT-5-mini as LLM, text-embedding-3-large as embedder

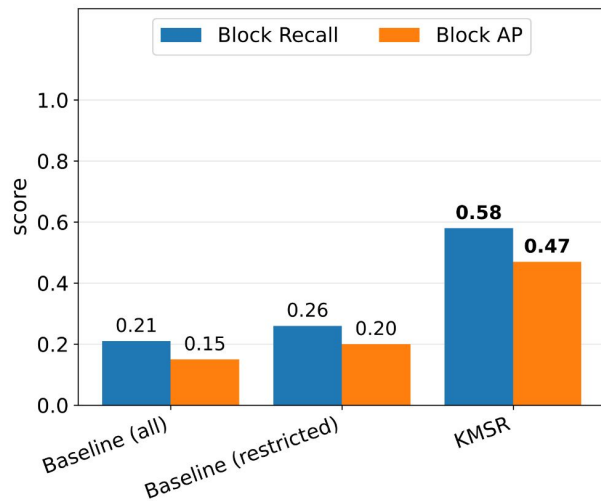
Results: Overview



Results: by Query Type



Flat queries



Block-structured queries



Key Findings and Limitations

- The retriever is not the lever
- considering structure in the splitting phase and reducing the hypothesis space by leveraging the Winfried knowledge base matters
- LLM reranking helps to mitigate noisy candidates
- The knowledge base is both the source of the gain and its ceiling:
→ the law can reclassify a circumstance as an autonomous offense. Therefore, only an up-to-date encoding of the code can solve this.

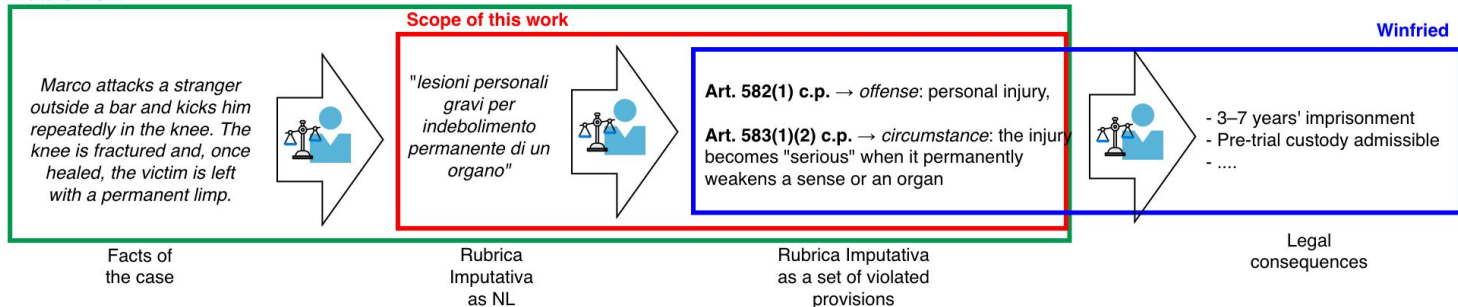
Conclusions and Future work

Research Question: Alongside LLMs, does injecting domain knowledge still add value?

YES! Especially when:

- the answer is compositional: a partial match is a wrong charge
- the decisive distinctions are legal, not lexical or semantic: similar wording is often misleading
- the candidate space is huge: knowledge restricts it to what the law admits

Future Work





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Thank You!

Questions?



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